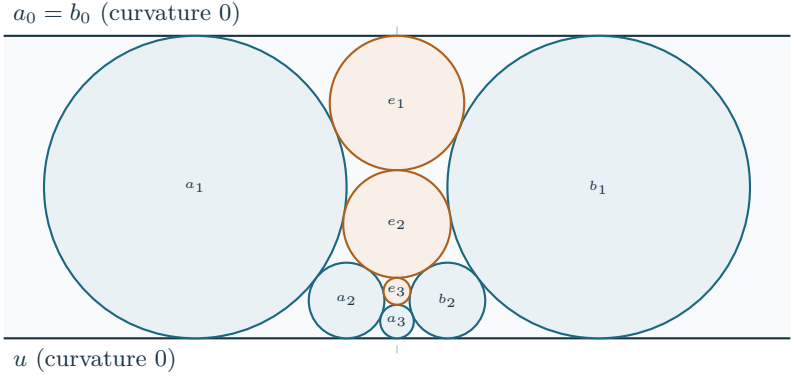
The exact seed for a uniform packing comparison

Reduced width $k = 4$: two marked lines, three circles on the symmetry axis.
The general proof compares a recurrence, not a list of numerical packings.



$$(a_0, a_1, a_2, a_3) = (0, 1, 4, 9), \qquad (e_1, e_2, e_3) = \left(\frac{9}{4}, \frac{45}{16}, \frac{45}{4}\right)$$

These labels denote curvatures; the drawn circle radii are their reciprocals.

The comparison used at every larger width

$$(1 + z_{j-1})(1 - z_j^2)(1 + z_{j+1}) = 1, \qquad z_0 = z_k = 0$$

$$k = 4 : \qquad (z_1, z_2, z_3) = \left(\frac{2}{3}, \frac{4}{5}, \frac{2}{3}\right)$$

With $v_j = \log(1 + z_j)$ the equation has a strictly convex right side.
Translated copies of this seed give $z_j \geq 2/3$ for every $k \geq 4$.